Unit 03: Project Life Cycle

Topics to be covered:

3.1 Introduction:

- Projects, by nature, are temporary with a definite start and end point.
- To achieve their goals successfully, they follow a sequence of well-defined phases, collectively known as the **Project Life Cycle**.
- These phases help project managers and teams organize work, manage risks, and track progress systematically.

3.2 Project Life Cycle – General:

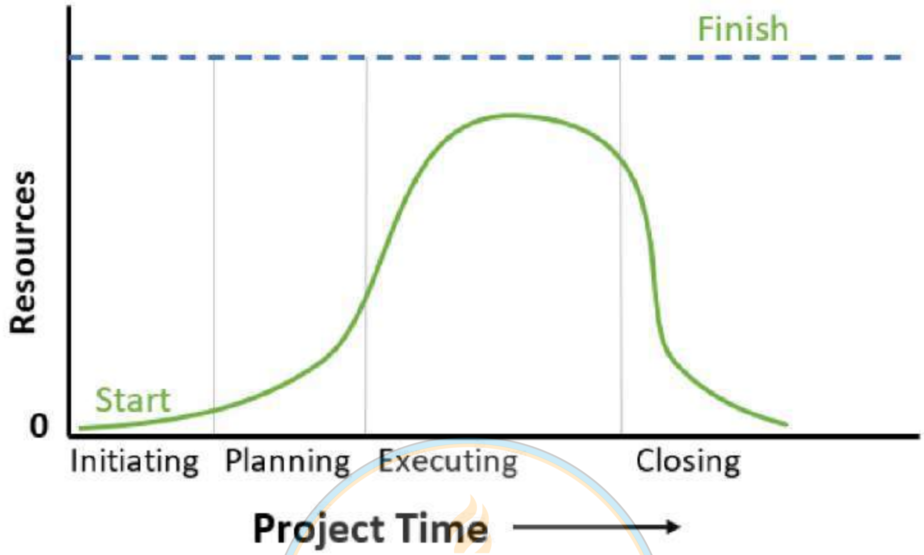
- The **Project Life Cycle** is the step-by-step process a project follows from start to finish.
- It helps in organizing work, managing time and budget, and ensuring success.
- It ensures everything runs smoothly from start to end.

Project Life Cycle have five main phases:

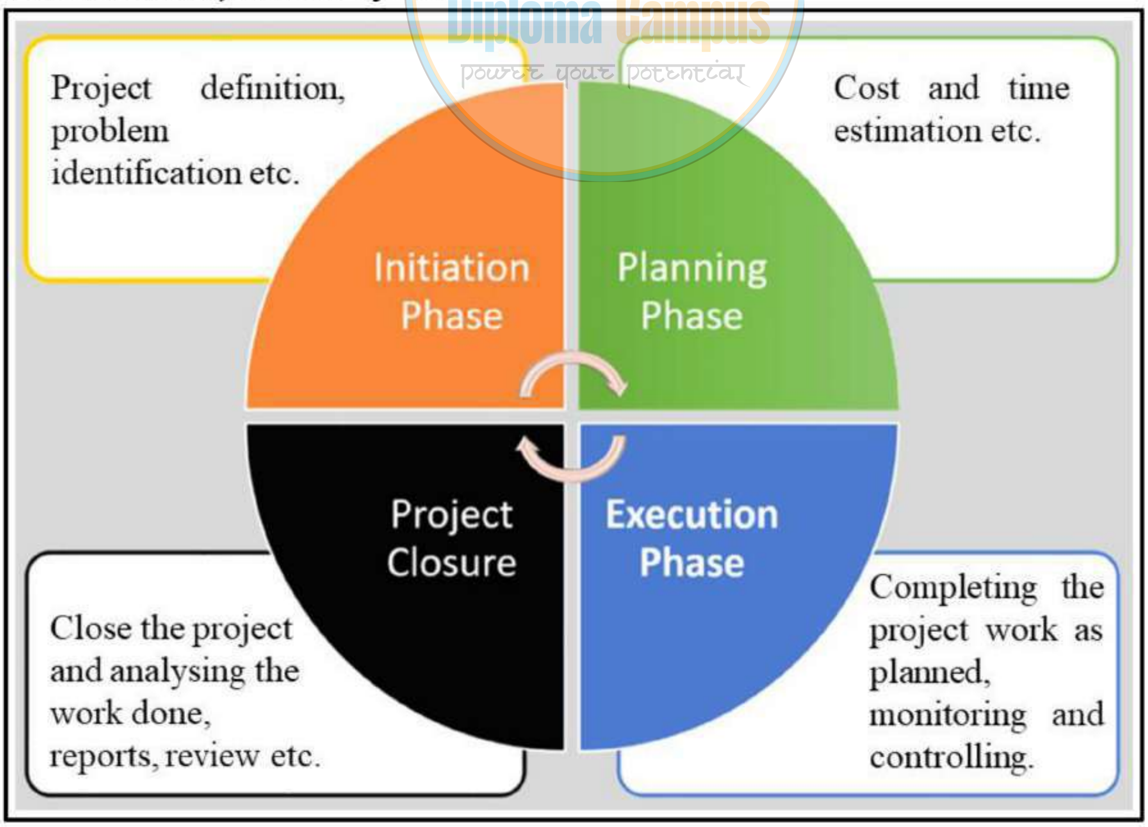
i. **Initiation** – Define the project goal and check if it is possible.
ii. **Planning** – Create a plan with a timeline, budget and team roles.
iii. **Execution** – Carry out the plan and complete tasks.
iv. **Monitoring & Controlling** – Track progress and fix any issues.
v. **Closure** – Finish the project, handover the deliverables to costumers, review the results and document lessons learned.

Project life cycle curve.

- Project life cycle curve is a parabolic pattern which indicates growth, maturity and decline of a project.
- This curve helps a project manager to analyse the state of project at any point of time.



Phases of Project Life Cycle:



3.3 Project Initiation and list the parameters:

i. Project Initiation:

- ❑ Project initiation is the first phase (stage) in managing a project.
- ❑ It sets the foundation for everything that comes next.
- ❑ It ensures the project is clear, realistic and ready to begin.

Parameters of Project Initiation:

- Understand what the project is about and why it is important.
- Check if the project is possible with the available resources, time and budget.
- Identify the people (stakeholders) involved and their roles.
- Get official approval to start the project.

3.4 Project Planning and list the parameters:

ii. Project Planning:

- ❑ Project planning is the second phase (stage) of project management.
- ❑ It involves creating a roadmap to guide the team on how to complete the project successfully.
- ❑ This phase ensures clarity, organization and preparation before work begins.

Parameters of Project Planning:

- Break the project into smaller, manageable tasks.
- Set clear timelines and deadlines.
- Allocate resources (people, money, tools) efficiently.
- Identify risks and prepare solutions.
- Establish communication plan (how communication will happen).

3.5 Project Execution and list the parameters:

iii. Project Execution:

- ❑ Project execution is the phase where the work is done as per the plan.
- ❑ It involves coordinating people, resources and tasks to complete project goals.
- ❑ This is the "action" phase of project management.

Parameters of Project Execution:

- Complete tasks and deliverables as per the plan.
- Ensure the team stays on fixed time and within budget.
- Maintain communication and address issues (problems) as they arise.
- Monitor progress to ensure quality and efficiency.

Project Monitoring & Control and list the parameters:

iv. Project Monitoring & Control:

- ❑ Project monitoring & control is the phase where you track the progress and performance of the project to ensure it stays on track.
- ❑ It helps identify issues (problems) early so corrective actions can be taken.
- ❑ It ensures project goals are achieved, within the planned timeline, budget and scope.

Parameters of Project Monitoring:

- Measure progress against the project plan.
- Ensure tasks are completed on time and within budget.
- Identify and resolve issues or risks early.
- Maintain quality standards.

3.6 Project Closure and list the parameters:

v. Project Closure:

- ❑ Project closure is the final phase of project management.
- ❑ It involves wrapping up all project activities, delivering the final product or service and ensuring all stakeholders are satisfied.
- ❑ This phase ensures the project is formally completed and lessons are captured for future projects.

Parameters of Project Closure:

- Deliver the project's final results to the client or stakeholders.
- Confirm all project tasks are completed.
- Release project resources (team members, tools and budgets).
- Document lessons learned and best practices.
- Celebrate the team's success!

Components of project planning. ***(March/April-2022)***

Following are some of the components of project planning:

i. Resource plan (Time, cost, men, material, machines etc.)
ii. Financial plan (Cost estimation)
iii. Quality plan
iv. Risk plan
v. Acceptance plan (Success or fail prediction)
vi. Communication plan
vii. Procurement plan (Purchase plan)

Need of project planning in project management. ***(Feb-March-2023)***

- Project planning is thinking before doing.
- Project Planning is all about designing effective policies and methods to complete project successfully.
- Reasons, for need of project planning are as follows:
  - To completely define all work to project team.
  - To reduce uncertainty.
  - To reduce risk.
  - To improve the efficiency of project (reduce wastages).
  - To obtain a better understanding of the project goals and objectives.
  - To provide a basis understanding on monitoring and controlling, project work.

Why project evaluation needed in project management cycle? ***(Feb-March-2023)***

- Project Evaluation is a step-by-step process of collecting, recording and organizing information about project results, including short-term and long-term project outcomes.
- Project evaluation provides answers to several questions such as:
  - Progress made.
  - Effective and efficient use of resources.
  - Desired output achieved.
  - Improvements to be made for better outcome.
  - Success factors
  - Whether the results justify the input etc.

Summarise project closure phase in project management. ***(Nov-Dec-2022-Makeup Exam)***

Project Closure:

- This is the final phase to do the official closure of the project.
- In this phase project deliverables (output) is handed over to the customers.
- Documents are handed over to the authority.
- Contracts are officially closed with venders and suppliers.
- Staffs are relived.
- Machines or equipment's are set free for the next project.
- A performance review is done at the end of this phase which will identify the positive and negative points of the project, which will help the project manager to undertake the next project.
- Major activities of the project closure phase are:
  - ✓ *Generating project closure report.*
  - ✓ *Handing over the project deliverables to customers.*
  - ✓ *Relieving the team members.*
  - ✓ *Conducting performance review for improvement.*

3.7 Project Risks

- Risk is defined as the possibility of an outcome being different from the expected outcome.
- Risk are uncertainties in any project, they can adversely affect the desired outcome of the project unless they are minimized.
- The risk is inherent in every activity of a project.
- The risk can only be minimised, it cannot be completely eliminated (Risk is inherent).
- All projects are exposed to various types of risks.

3.8 Types of Risks: Illustrations

Types of project risks are as follows.

i. ***Technical Risks:*** Risk due to changes in technology or change in technical specification (new technology).
ii. ***Social Risks:*** Social risks refer to risks arising from changes in the needs and preferences of customers. Lack of necessary natural resources, labour tensions and social movements against the project are some social risks.

iii. ***Economic Risks:*** Economic risks refer to an increase in the rate of inflation, changes in the economic policies of governments.

iv. ***Political Risks:*** Nationalisation or privatisation of a particular industry, political instability, and trade restriction are some examples of political risks. The project manager must ensure that the project does not go against the political interests of the country.

v. ***Production Risks:*** Production risks refer to the shortage of required raw materials, sudden breakdown of key machinery and huge installation and maintenance costs.

vi. ***Marketing Risks:*** Marketing risks refer to fail of project due to change in market demand also difficulties in distribution.

vii. ***Financial Risks:*** Financial risks refer to bad debts (loan), change in the interest rate, wrong choice of investments and mistakes in the accounting.

viii. ***Human Risks:*** Human risks refer to the sudden demise (death) of key employee, limited availability of skilled employees, inter-group politics, etc.

Role of project manager to minimize risk in a project.

Project manager plays an important role in minimising project risk.

- He holds the overall control of the project and responsible for its completion.
- He must involve in planning, monitoring, directing and leading team members and makes sure that project goals are reached in time, cost and quality.
- He must maintain a project dairy to record project activities and progress.
- He must reduce the wastage of resources (time, money, men, materials, machine etc.)
- He must be flexible and adoptable to any situation.
- He must properly communicate the project progress to all team members and stakeholders.
- Project manager should be confident, verbally fluent and must be a specialist in communication.
- He must be able to identify the problems ahead.
- He must be able to maintain a proper balance between time and money.
- He must take strong decisions and it must be accepted by his team members.

Risk Management:

- Risk management is a process of identifying potential risk in a project and controlling it with some actions.

The risk management process involves the following:

- Identifying critical and non-critical risks.
- Documenting each risk in depth by completing risk forms.
- Creating a record of all risk and inform the management of their severity.
- Taking actions to reduce risks.
- Reducing the impact of risk on the project by following the best practice and best methods.

3.9 Cost Overruns:

- Cost Overruns are the condition where the project does not complete within the given budget or cost.
- This may be the result of poor planning and poor management.
- Proper planning, controlling, tracking and monitoring with review system can always avoid cost overrun.

Following may the reasons for cost overrun:

- Poor Management.
- Poor Administration.
- Poor Planning.
- Poor cost estimation.
- Poor Resource Plan.
- Poor Quality Plan.
- Poor Communication Plan.
- Poor risk Management.
- Poor skilled Team Members.
- Poor skilled workers.
- Poor materials from suppliers.
- Fail to make work breakdown structure.
- Fail to track project spending.
- Fail to review similar projects in past.
- Re-work or re-assembly.
- Errors in project design.
- Unplanned expansion of the project scope.
- Wrong choice of equipment.
- Time overrun.

3.10 Time Overruns:

- It is the condition where the project does not complete within the scheduled time (given time).
- This may be the result of poor planning and poor time management.
- Proper planning, controlling and monitoring with review system can always avoid time overrun.

Following may the reasons for time overrun:

- Poor Management.
- Poor Administration.
- Poor Planning.
- Poor Time estimation.
- Poor Resource Plan.
- Poor Quality Plan.
- Poor Communication Plan.
- Poor risk Management.
- Poor skilled Team Members.
- Poor skilled workers.
- Delay of materials from suppliers.
- Political interference.
- Fail to make work breakdown structure.
- Fail to track project spending.
- Fail to review similar projects in past.
- Re-work or re-assembly.
- Errors in project design.
- Unplanned expansion of the project scope.
- Improper project team building.
- Wrong choice of equipment.
- Changing project objective or events.
- Delay in start-up, execution and closure.
- Delay in undertaking dependent project stages.
- Poor time table or poor follow up.
- Possible interruption from public.
- A change in the scope of the project.
- Delays in starting and executing some of the project activities.
- A delay in one project, results in delays in subsequent projects.
- Use of out-dated technology.
- Cost overrun.

3.11 Case Studies

Case Study 2: Assume you are organizing a 5-day educational trip.

Requirements:

1. There are 30 students in the trip and 2 staffs accompanying them.
2. Mode of transport to be decided.
3. Food arrangements has to be made for the students and staff.
4. Accommodation has to be made for the students and staff.
5. Emergency (First-Aid) Medical arrangements has to be made.

Tasks:

- Prepare the detailed WBS.
- Prepare various phases of the project according to the project life cycle and duration.
- Prepare the project dairy template.
- Prepare the project plan with risks involved and management of the risk.
- Estimate the budget for the entire activity.
- Discuss the reasons of Cost and Time overrun if project is delayed.
- Present it in a seminar, with each group getting 5-10 minutes to present their idea.

Case Study 2 Solution:

Case Study 2: Organizing a 5-Day Educational Trip

1. Introduction

An educational trip for 30 students and 2 staff members is planned for five days. The project involves organizing transportation, food, accommodation, and emergency medical arrangements while ensuring cost and time efficiency.

2. Work Breakdown Structure (WBS)

Educational Trip Planning

- 1.1 Pre-Trip Planning
  - Define objectives & budget
  - Select destination & itinerary
  - Obtain approvals
  - Risk assessment
- 1.2 Transportation
  - Select transport mode (bus/train/flight)
  - Arrange bookings
  - Confirm pickup/drop points

    - Plan backup transport
- 1.3 Accommodation
    - Identify & book lodging
    - Allocate rooms
    - Ensure security
- 1.4 Food Arrangements
    - Select meal plans
    - Coordinate with food vendors
    - Arrange for dietary needs
- 1.5 Emergency Medical Arrangements
    - First-aid kits & medical contacts
    - Nearest hospitals identified
    - Staff assigned for emergencies
- 1.6 Activities & Supervision
    - Educational sessions
    - Recreational activities
    - Student supervision
- 1.7 Post-Trip Closure
    - Collect feedback
    - Financial reconciliation
    - Final report

3. Project Life Cycle & Phases

Phase	Duration	Key Activities
Initiation	1 week	Define objectives, budget, approvals
Planning	2 weeks	Itinerary, bookings, risk planning
Execution	5 days	Trip execution, monitoring
Monitoring & Control	5 days	Safety, logistics tracking
Closure	1 week	Feedback, financial reconciliation

4. Project Diary Template

Educational Trip Project Diary

- Date:
- Day Number:
- Location:
- Activities Planned:
- Challenges Faced:

- Risk Incidents & Mitigation:
- Student & Staff Feedback:
- Lessons Learned:
- Signatures:

5. Project Budget Estimation

Expense Category	Estimated Cost (per person) (INR)	Total Cost (for 32 people) (INR)
Transportation	₹ 4,000	₹ 1,28,000
Accommodation (5 days)	₹ 8,000	₹ 2,56,000
Food (3 meals/day)	₹ 3,500	₹ 1,12,000
Medical & Emergency	₹ 1,500	₹ 48,000
Miscellaneous	₹ 2,000	₹ 64,000
Total Estimated Cost	₹ 19,000	₹ 6,08,000

6. Cost & Time Overrun Analysis

Reasons for Cost Overrun:

1. Unexpected price changes (transport, accommodation).
2. Medical emergencies requiring extra expenses.
3. Additional activities or extended stay.

Reasons for Time Overrun:

1. Transport delays (traffic, weather issues).
2. Unplanned activities extending schedule.
3. Delays in check-ins, food service.

Mitigation Strategies:

- Buffer time in schedule.
- Contingency budget (10-15% extra).
- Pre-arranged alternative transport/lodging.

7. Seminar Presentation Plan

Presentation Structure (5-10 Minutes)

1. Introduction (1 min) – Purpose & objectives of the trip.
2. Project Plan (2 min) – WBS, life cycle phases.
3. Logistics & Budget (3 min) – Transport, food, accommodation, costs.
4. Risk Management (2 min) – Cost/time overruns & mitigation.
5. Conclusion (2 min) – Key takeaways & Q&A.

Question bank (05 to 10 Marks):

1. Explain Project Life Cycle.
2. Explain Project Life Cycle Curve with the help of a diagram.
3. Explain phases of Project Life Cycle with help of a neat diagram.
4. Explain Project Initiation and list its parameters.
5. Explain Project Planning and list its parameters.
6. Explain Project Execution and list its parameters.
7. Explain Project Monitoring & Control and list its parameters.
8. Explain Project Closure and list its parameters.
9. Explain Project Risk and list the different types of project risks.
10. Define Technical risk.
11. Define Social risk.
12. Define Economic risk.
13. Define Political risk.
14. Define Production risk.
15. Define Marketing risk.
16. Define Financial risk.
17. Define Human risk.
18. What are the roles of Project manager to minimize risk in a project?
19. Explain Risk management.
20. Explain Cost overrun and list the possible reasons for Cost overrun.
21. Explain Time overrun and list the possible reasons for Time overrun.